

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Final Report

Submitted by :

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Institution :

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Duration :

7-Day Capstone Project

Tools Used :

Python, Pandas, NumPy, Matplotlib,
Seaborn, Plotly, SQLite, Power BI, VS Code

Problem Statement

Mutual fund investors require accurate and meaningful insights to make informed investment decisions. Large volumes of financial data often make it difficult to identify trends, evaluate performance, and understand investor behavior.

The objective of this project is to transform raw mutual fund data into actionable insights through data analytics and visualization techniques.

Objectives :

- Build an ETL pipeline for mutual fund datasets.
- Store structured data using SQLite.
- Perform exploratory data analysis.
- Compute performance metrics for mutual funds.
- Conduct advanced risk analysis.
- Analyze investor behavior.
- Develop an interactive Power BI dashboard.
- Generate recommendations based on fund performance.

Data Source :

Ten datasets were used in this project:

1. Fund Master Data
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category-wise Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

These datasets contain information regarding fund performance, investor transactions, asset allocation, and market benchmarks.

ETL Design :

The ETL process consists of:

Extraction

CSV datasets were collected and imported using Python.

Transformation

Data cleaning operations included:

- Handling missing values
- Converting date columns
- Standardizing formats
- Validating data consistency

Loading

Processed datasets were loaded into a SQLite database for structured querying and analysis.

Database Design :

SQLite was used as the backend database.

The database schema contains tables corresponding to each dataset:

- . fund_master
- . nav_history
- . aum_by_fund_house
- . monthly_sip_inflows
- . category_inflows
- . industry_folio_count
- . scheme_performance
- . investor_transactions
- . portfolio_holdings
- . benchmark_indices

The schema enables efficient storage and retrieval of financial data.

Exploratory Data Analysis :

Exploratory Data Analysis was performed to understand trends and patterns within the datasets.

Several visualizations were created using Matplotlib, Seaborn, and Plotly.

Major analyses included:

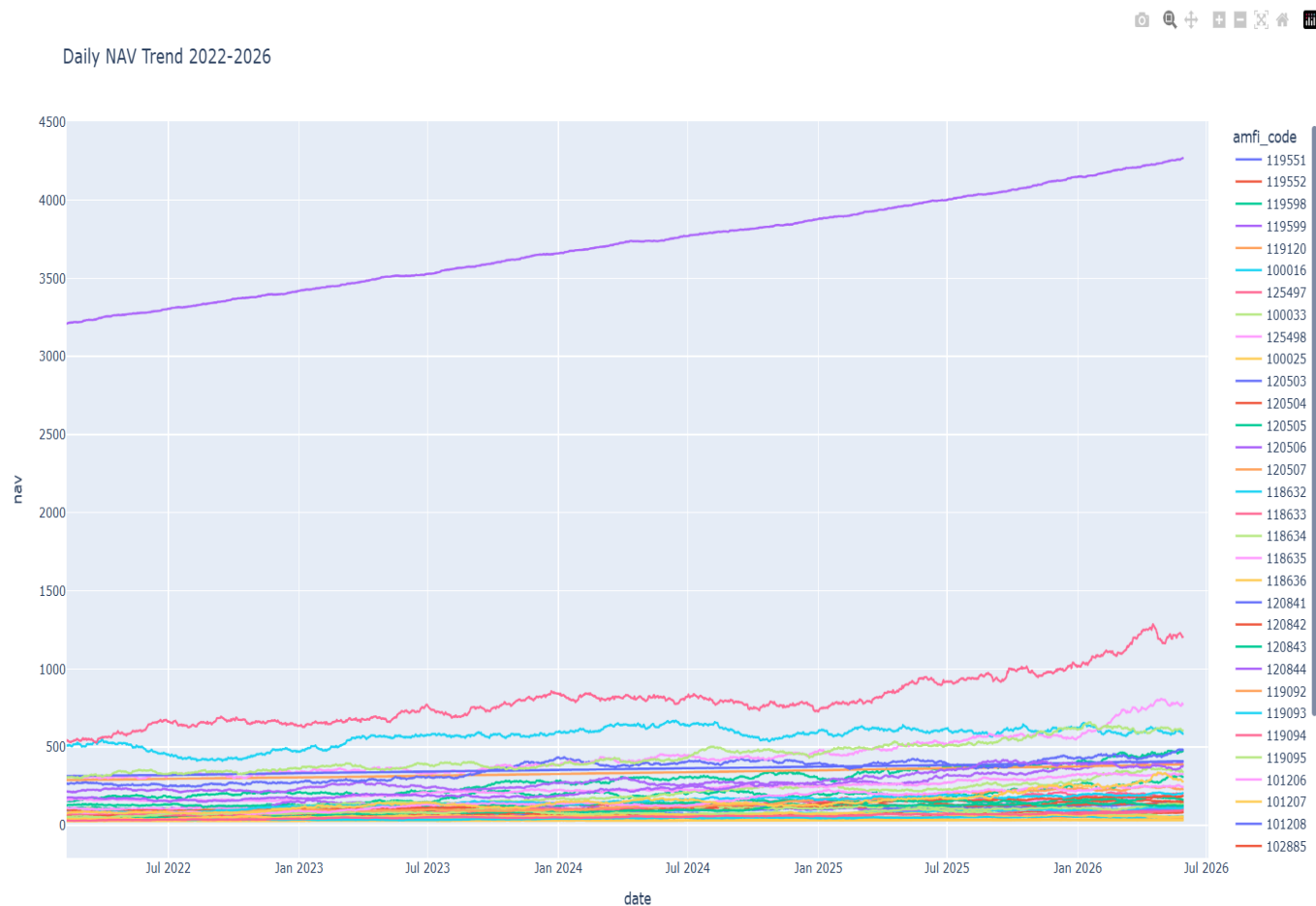
- . NAV trends
- . AUM growth
- . SIP inflow trends
- . Category inflows
- . Investor demographics
- . Geographic distribution
- . Folio count growth
- . Correlation analysis
- . Sector allocation

NAV Trend Analysis :

Daily NAV trends from 2022–2026 were analyzed for multiple mutual fund schemes.

The analysis captured:

- Long-term growth patterns
- Market corrections
- Bull market phases

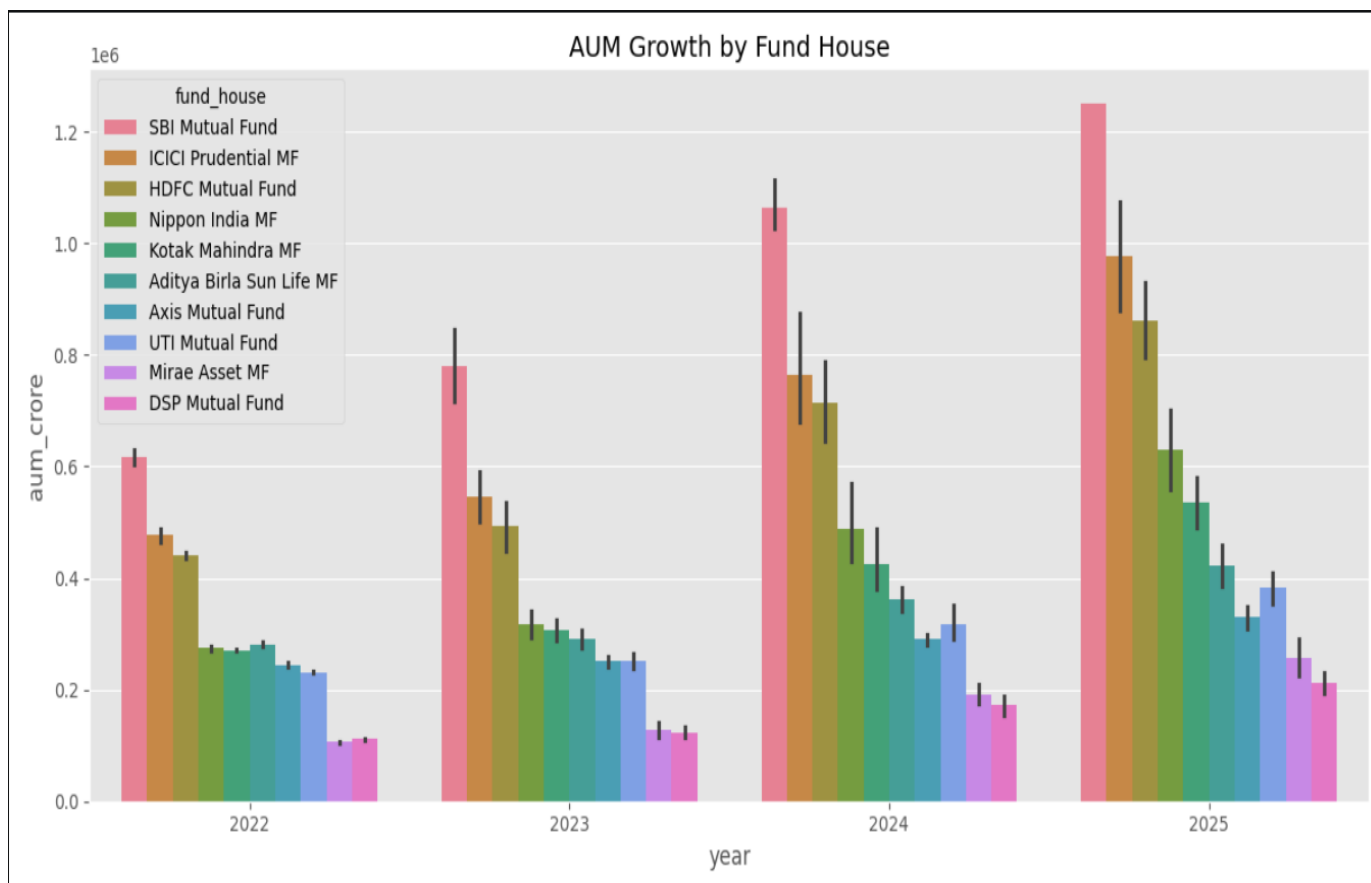


AUM Growth Analysis

Asset Under Management growth was analyzed across major fund houses.

Important observations:

- Significant industry expansion occurred between 2022 and 2025.
- SBI Mutual Fund maintained dominant market share.



Investor Analytics

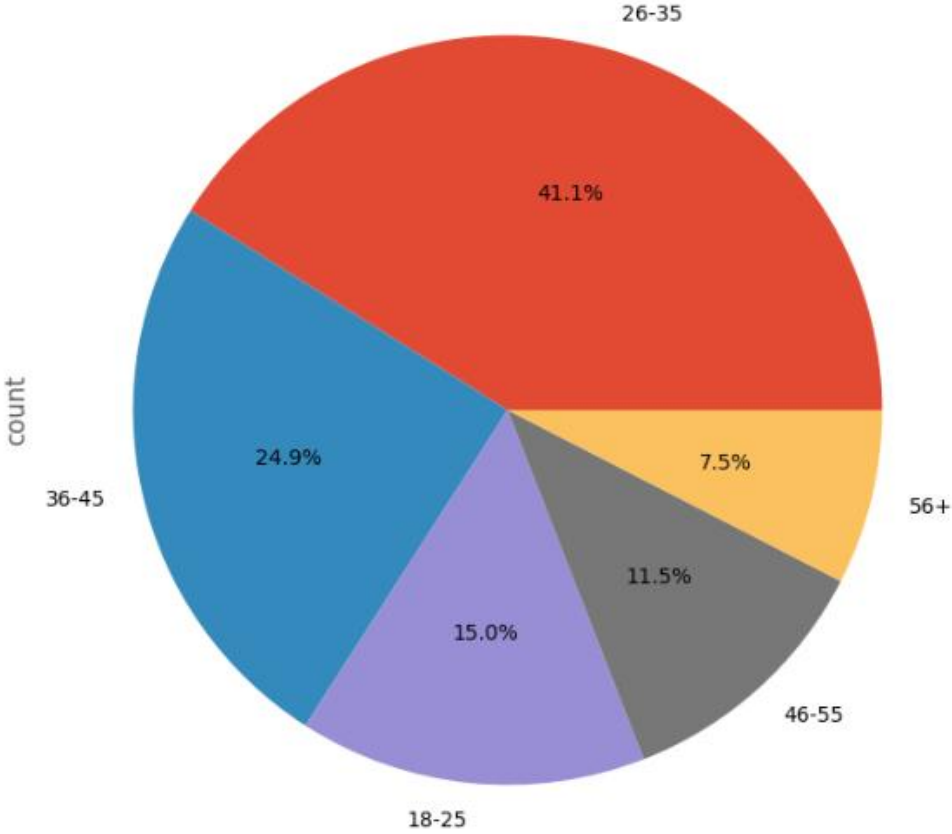
Investor behavior was studied using transaction data.

The analysis included:

- Age group distribution
- Gender split
- State-wise investments
- City-tier segmentation

These insights help understand investor participation across demographics.

Investor Age Group Distribution



Performance Analytics

Performance metrics were computed for all schemes.

Metrics included:

- . Daily Returns
- . CAGR
- . Sharpe Ratio
- . Sortino Ratio
- . Alpha
- . Beta
- . Maximum Drawdown

These measures enabled comparison of risk-adjusted performance.

Advanced Analytics

Advanced techniques were applied to evaluate downside risk and investor behavior.

Analyses performed:

- . Historical VaR
- . Conditional VaR (CVaR)
- . Rolling Sharpe Ratio
- . Investor Cohort Analysis
- . SIP Continuity Analysis
- . Sector HHI Concentration
- . Fund Recommendation System

These analyses provided deeper insights into risk and portfolio diversification.

Dashboard Development :

Power BI was used to create a four-page interactive dashboard.

Dashboard pages include:

Industry Overview :

- . Total AUM
- . SIP inflows
- . Folio counts
- . AUM trends

Fund Performance :

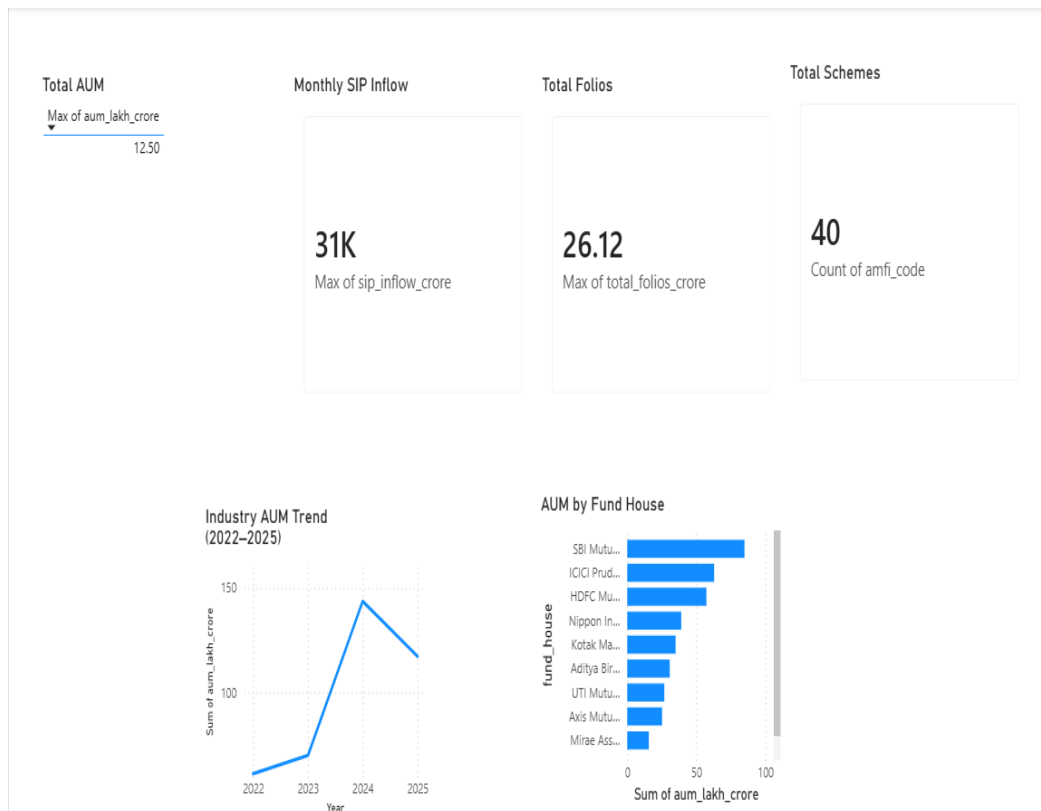
- . Risk-return analysis
- . Fund scorecards
- . Benchmark comparison

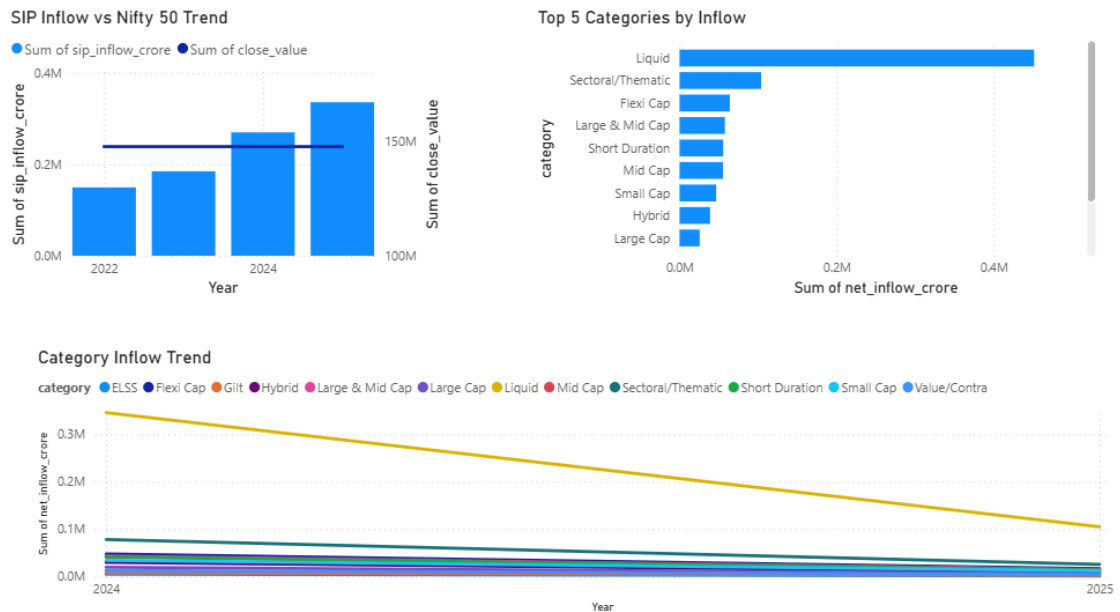
Investor Analytics :

- . State-wise transactions
- . Age group distribution
- . Monthly transaction trends

SIP and Market Trends :

- SIP inflows
- Category-wise inflows
- Market index trend





Key Findings :

- Mutual fund industry experienced continuous growth.
- SIP inflows reached record highs during the study period.
- Risk-adjusted metrics identified top-performing schemes.
- Investor participation varied across states and age groups.
- Portfolio concentration differed significantly among funds.

- Advanced risk measures highlighted downside exposure.

Limitations :

- Historical data availability was limited.
 - Market conditions can change over time.
 - External macroeconomic events were not incorporated.
 - Future returns cannot be guaranteed based on historical performance.
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Recommendations :

- Investors should focus on risk-adjusted returns rather than absolute returns.
- Portfolio diversification should be encouraged.
- SIP continuity can improve long-term wealth creation.
- Fund selection should align with risk appetite.

- . Continuous monitoring of portfolio performance is essential.

Conclusion :

The Bluestock Mutual Fund Analytics Capstone Project successfully implemented an end-to-end analytics framework for the mutual fund industry. The project integrated data engineering, exploratory analysis, performance measurement, advanced risk analytics, and business intelligence techniques.

The resulting insights and dashboards provide a practical foundation for informed investment decision-making and demonstrate the application of data science in financial analytics.